

HACKATHON

BUILD BOLD SOLUTIONS.

CREATE REAL-WORLD IMPACT.

INNOVATE

SOLVE

TRANSFORM

IMPACT

Hackathon name: Hackathon Dragon's Den

Solution name: AI Operations Intelligence Agent

Team name: InsightForge-AI

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Idea. Build. Scale.

LET'S BUILD TOMORROW,
TOGETHER.

AI Operations Intelligence Agent

InsightForge-AI | Dragon's Den pitch

CORE MESSAGE

Dashboards show the number. InsightForge detects what changed, investigates why, identifies who is impacted, and tells you what to do next.

Not another dashboard

AI root-cause investigation

Evidence-backed actions

Scenario used throughout: 25,000 customer records | Complaint cases: 1,200 to 1,650 | RCA: 4 hours to 5 minutes



Business Problem

The core challenge our AI Operations Intelligence Agent is solving

Main business problem

Utility teams can see operational KPIs changing in existing dashboards, but they cannot quickly understand the real business reason, impacted customer groups, and priority action required for each user group.

Key challenges

- Root-cause visibility is fragmented across multiple operational processes and systems.
- Business teams spend too much time interpreting data before action can begin.
- Different analysts or teams may reach different conclusions from the same KPI movement.
- Operational decisions become reactive instead of proactive.
- Delayed understanding leads to reactive customer and operational management.

Business impact

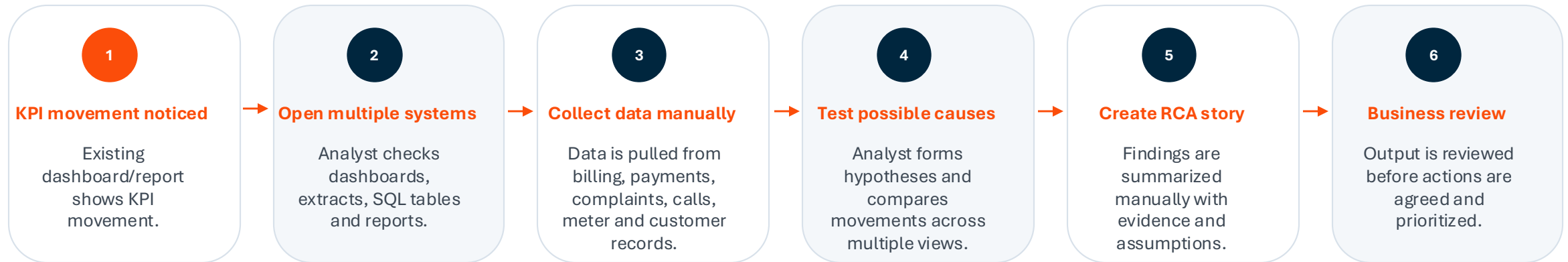
- Customer-impacting issues can escalate before the true driver is identified.
- High-risk customers, regions, segments or processes may not be prioritized early enough.
- Leadership receives KPI movement updates without clear evidence or action confidence.
- Resources are not always focused on the issues with the highest business impact.

What needs to change: Move from “KPI changed” to integrated report-level insight: why it changed, who is impacted, what evidence supports it, and what action each user group should take next.

Old Way of Analysis

How KPI movement was investigated before the AI Operations Intelligence Agent

Manual investigation journey: fragmented, slow and analyst-dependent



Why the old way creates friction

- KPI deviations may not be detected early enough to enable timely investigation
- Root-cause evidence is spread across different operational datasets.
- Analysis depends heavily on analyst experience and local knowledge.
- The same KPI movement can lead to different explanations across teams.
- Business action starts after manual data stitching and review.

What this means for business users

- Stakeholders see the issue before they understand the business reason.
- Customer-impacting trends may worsen while teams investigate.
- Operational resources may not be focused on the highest-impact problem.
- Leadership receives delayed insight without a consistent evidence trail.

Transition to the new way: InsightForge replaces manual data stitching with an AI-guided investigation that explains why the KPI changed, what evidence supports it, and what action should be taken next.

The Hidden Cost of Delayed KPI Investigation

Traditional investigation often starts after the deviation has already created impact.

BUSINESS IMPACT

Additional complaint cases
+450

Service-cost exposure
£250K+

- Without root cause, complaint resolution can be delayed or generic.
- Leadership needs evidence, not just a KPI movement.
- Potential revenue leakage due to delayed corrective actions.
- Reputational damage caused by repeated service issues.
- Loss of customer trust and retention.

CUSTOMER IMPACT

Vulnerable customers
800

Affordability flagged
15%

- Vulnerable and repeat-contact customers need earlier support.
- Potential complaint increase is 12% if root cause is left unresolved.
- Customer impact makes the use case more than operational efficiency.
- Customers may switch suppliers following poor experience.

WHY STAKEHOLDERS CARE

- Service-cost exposure can be quantified.
- At-risk customer records can be prioritized.
- Customer support can be targeted.
- Action can be tracked against the true driver.

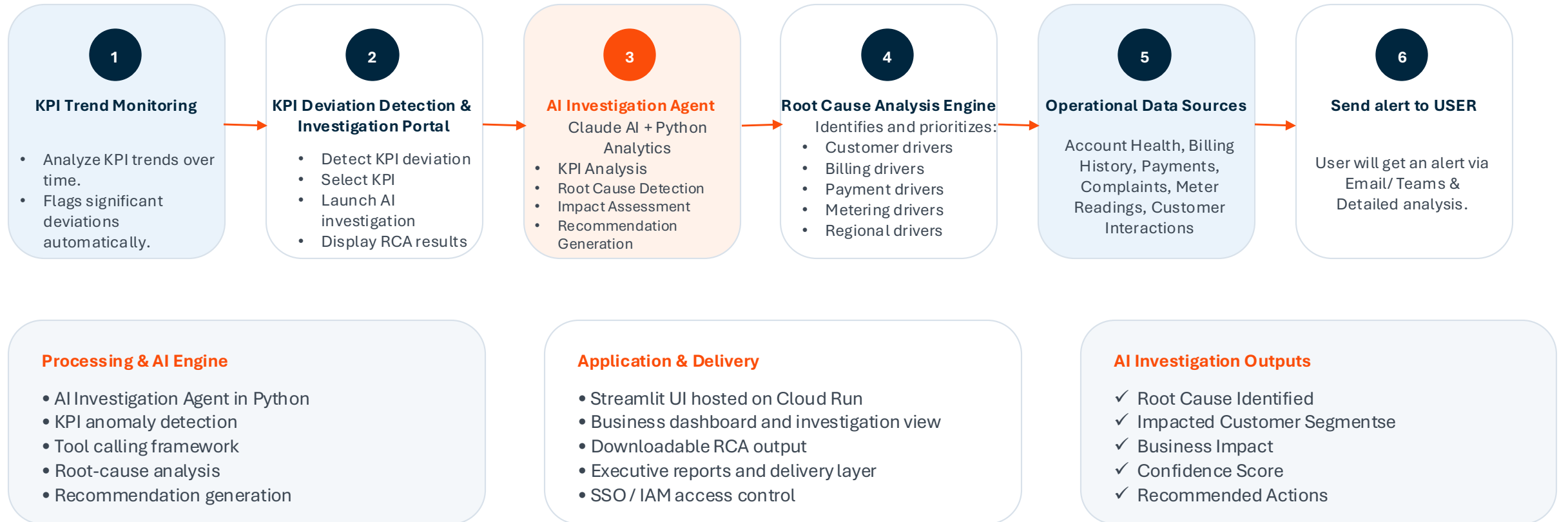
POSITIONING FOR DRAGON'S DEN

InsightForge is not a standalone product; it integrates with existing dashboards/reports to help analysts and managers identify root cause faster, quantify who is impacted, and generate action items for business, consumer-service and sales/operations users.

New Way of Analysis

AI-driven, evidence-based and action-oriented

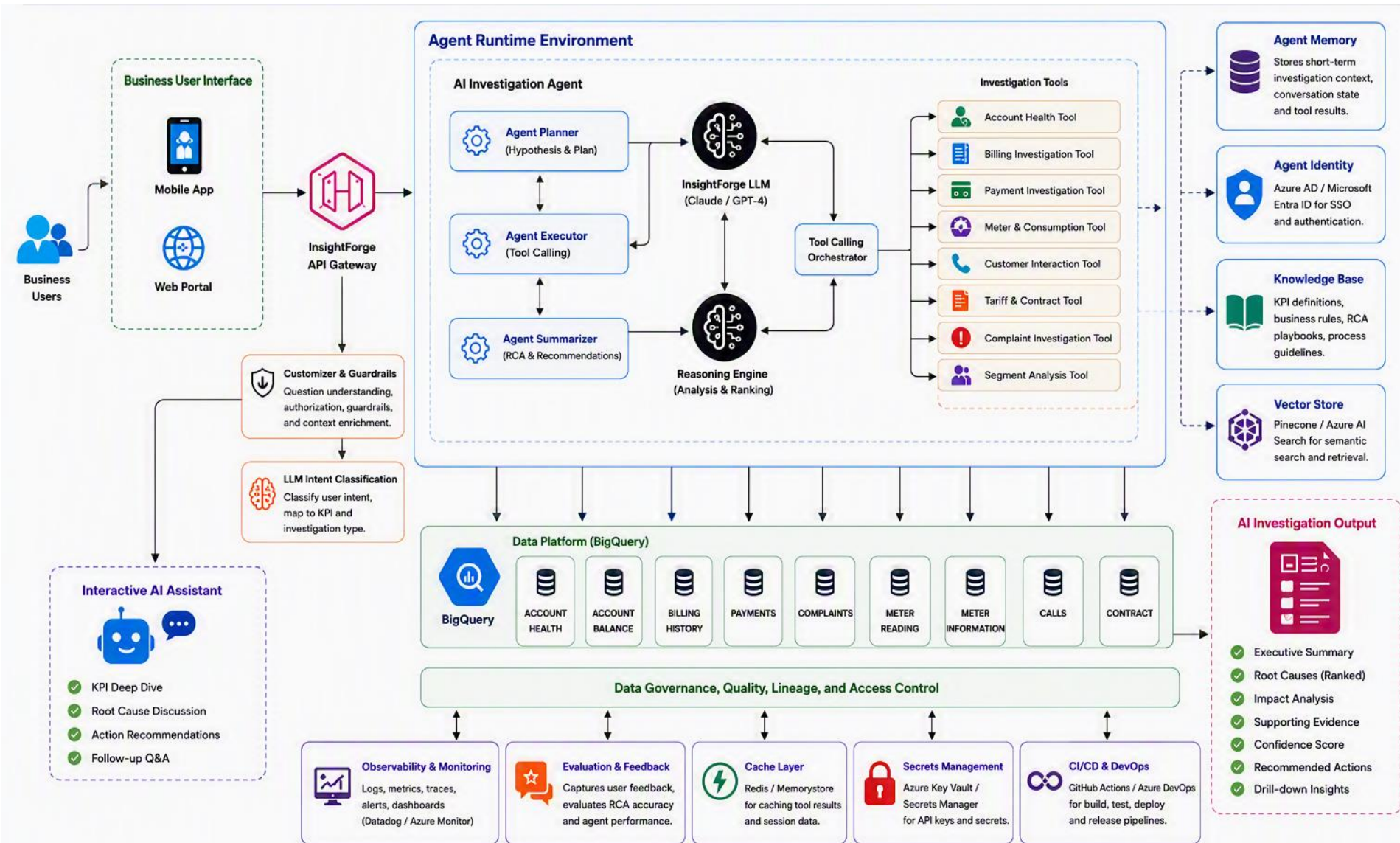
Technical Architecture: From business question to AI-led root-cause action



Architecture outcome: KPI deviation is investigated inside the existing report/dashboard flow, root causes are identified across operational datasets, and end users receive role-based actions to reduce customer and revenue impact.

High-Level Architecture: InsightForge Inside Existing Dashboards

Detailed integrated architecture view showing dashboard integration, AI investigation, tool calling and role-based outputs.



- Shows InsightForge as an embedded AI layer, not a standalone product.
- Clarifies integration with existing dashboards such as Power BI, Tableau and reporting portals.
- Connects business users to AI investigation, tool calling, RCA and recommended actions.
- Highlights role-based outputs for business, consumer-service and sales/operations users.
- Makes the architecture easier for judges and stakeholders to understand in one view.

Slide purpose:

One-view architecture for explaining how InsightForge fits into the existing reporting ecosystem.

Feasibility to Incorporate in Existing Systems

Designed as an additional intelligence layer on top of current reporting and analytics capabilities.

Implementation principle: reuse the existing reporting setup and add AI-led investigation on top.

1

Reuse KPI logic

Existing KPI calculations and reporting logic can remain in place.

2

Connect to BigQuery

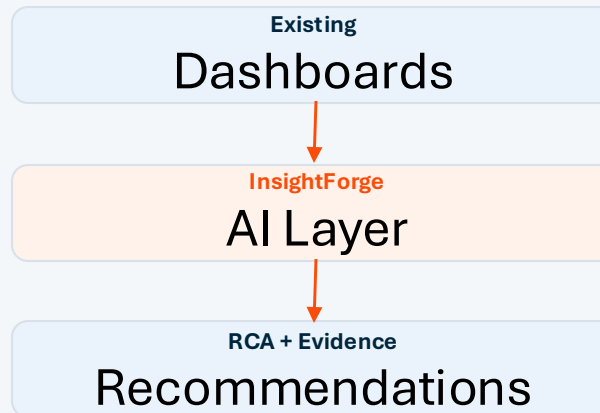
Agent connects to approved reporting datasets rather than rebuilding source systems.

3

Use existing domains

Complaint, billing, meter and customer data can be accessed through focused tools.

How it fits into current estate



Why it is feasible

- Cloud Run can host the Streamlit UI and AI services.
- OpenAI / Claude API integration supports the agent layer.
- Tool calling framework limits the agent to targeted metric retrieval.
- PoC can start with Complaint Rate before expanding to more KPIs.

Judge takeaway:

The idea is feasible because it reuses existing reports, data and KPI logic, while adding an integrated AI investigation layer that turns KPI movement into RCA and role-based action items.

Why InsightForge Is Different

An integrated intelligence layer that turns dashboard trends into evidence-backed actions.

InsightForge is not a standalone product. It integrates with existing dashboards and reports.

1 Existing Dashboard / Report

Shows KPI movement, historical trends and performance change.

2 InsightForge AI Layer

Investigates why the KPI moved using dynamic tools, evidence and confidence scoring.

3 Role-Based Action Items

Recommends next-best actions tailored to business, consumer-service and sales/operations users.

Dashboard / Automation

- Dashboard shows KPI movement and trends.
- Automation executes fixed rules for known scenarios.
- Root-cause explanation still requires manual investigation.

InsightForge Intelligence

- Chooses investigation tools dynamically based on KPI context.
- Compares evidence across multiple drivers before concluding.
- Provides evidence points, confidence score and recommended actions.

Action Items by User Type

- Business users: root-cause summary and priority drivers.
- Consumer-service users: customer-impact and support actions.
- Sales/operations users: segment, region and follow-up actions.

Judge-friendly message:

Dashboards show what changed. InsightForge integrates with those dashboards to explain why it changed and generate action items for each business user group.

Compliance, Security & Governance

Controlled access model: the agent retrieves only the metrics required for investigation, not unrestricted raw data.

Compliance position: metrics are exposed through governed tools, while data remains inside approved reporting layers.

1

Data access control

- Read-only access to approved reporting datasets.
- RCA uses focused tool outputs, not full raw-data exposure.
- BigQuery remains the governed reporting layer.

2

Identity & credentials

- IAM supports role-based access control.
- Secret Manager stores API keys and credentials.
- SSO / IAM can control portal access.

3

Output governance

- Agent outputs evidence, drivers and recommendations.
- Business users see summarized RCA metrics and role-based action items.
- Existing governance controls remain in place.

Why this should not create a major compliance issue

The AI agent is not designed to expose complete operational datasets. It calls predefined tools and returns limited metrics, evidence and RCA summaries needed to explain KPI movement.

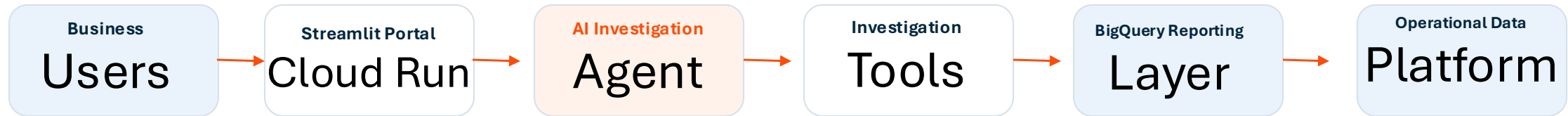
Governance message

InsightForge works inside the client governance model: governed datasets, role-based access, secure credentials and investigation outputs rather than uncontrolled data exposure.

Deployment Pattern

Cloud-native deployment pattern for PoC, pilot and enterprise rollout.

Deployment flow: Streamlit portal on Cloud Run, AI agent services, governed BigQuery data and operational data platform.



Environment

- GCP cloud-native deployment.
- Cloud Run for Streamlit and AI Agent.
- BigQuery reporting layer.
- Dev, UAT and Production environments.
- Auto-scaling serverless architecture.

Rollout approach

- Phase 1: Complaint Rate PoC.
- Phase 2: Business pilot with Operations, Billing and Customer Service.
- Phase 3: Enterprise rollout with SSO and governance.
- Phase 4: Multi-KPI expansion.

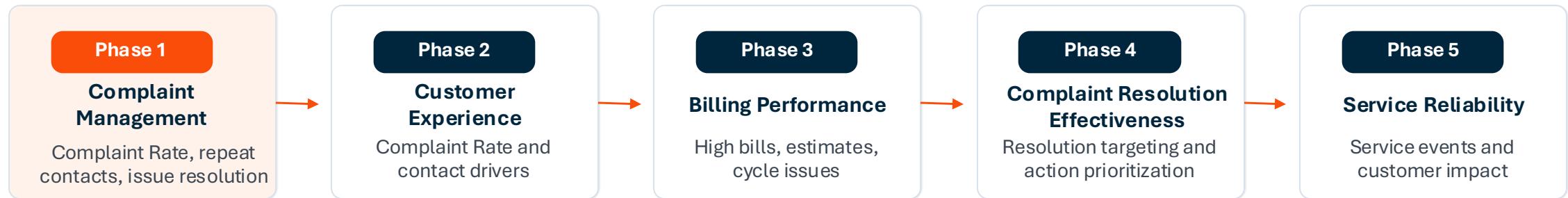
Ownership & support

- Analytics team owns KPI logic and validation.
- Data Engineering supports BigQuery and tool optimization.
- Cloud team supports infrastructure and IAM.
- Business owners validate RCA usefulness.

Deployment takeaway Start with a focused complaints PoC integrated into the existing report/dashboard, prove value with business users, then scale across additional operational KPIs.

Revenue & Business Growth

Start with Complaint Rate. Expand into KPI packs without retraining a new model.



WHY THIS CREATES BUSINESS GROWTH

- The same agentic investigation pattern can support multiple operational KPIs.
- New KPI packs require new tools/prompts, not new ML model training.
- Value increases as more teams use the same evidence-backed decision framework.

COMMERCIAL NARRATIVE

- Start small with one strong use case.
- Prove time saved and decision impact.
- Expand across domain-specific KPI packs.
- Position as an integrated operations intelligence layer inside existing dashboards/reports.

KPI packs
1 -> 5

Connected
tables
9

History
24
months

Transactions
250K+

Real-World Evidence: Cost of Delayed Complaint Resolution

Customer Impact

OVO Complaint Handling Case (2024)

- **1,395 customers impacted** by complaint-handling failures.
- **706 vulnerable customers affected.**
- Some complaints remained unresolved for **up to 18 months.**
- Demonstrates how delayed issue identification can directly impact vulnerable customers.

❑Source: Ofgem (2024)

Consumer Satisfaction

- Only **44% of customers** were satisfied with how their complaint was handled.
- More than half of complainants reported dissatisfaction with complaint resolution.

❑Source: Ofgem Energy Consumer Satisfaction Survey (2025)

Unresolved Customer Issues

- **34% of customers** reporting smart-meter issues stated that their problem was still unresolved or under investigation.

❑Source: Ofgem (2024)

Source: Ofgem Survey (2024)

Business Impact

Financial Penalties

- OVO paid **£378,512 directly to affected customers.**
- Additional **£2 million contribution** made to the Energy Industry Voluntary Redress Scheme.
- Total remediation cost exceeded **£2.37 million.**

❑Source: Ofgem Enforcement Action

Customer Churn & Revenue Risk

- **1.2 million energy customers** were estimated to be **£166 million worse off** after abandoning attempts to resolve issues with their supplier.

❑Source: UK Government consultation response

Operational Cost

- Customers spent an estimated **13.9 million hours** dealing with energy-service issues.
- Equivalent to roughly **1,600 years of customer time lost.**

Why this matters?

- Delayed complaint resolution creates customer dissatisfaction, regulatory exposure, reputational damage, and financial loss. Integrated Operation Intelligence AI helps identify root causes earlier and provides role-based action items before customer and business impacts escalate.



Thank you

InsightForge Team